IN4MATX 133: User Interface Software

Lecture 10:
Components in Angular

Announcements

- If you plan to start on A3 over the weekend, see Slack for configuration
- We're working on a "Plan C" and "Plan D" as well, potentially more updates over Slack soon

Announcements

- Discussion Friday (2/6) will offer more depth into Angular
- You're will soon be over the hardest part of the learning curve in this course
- And for me, we're starting to reach the more fun parts
- Start to take a look at A3!
- #assignment3 on Slack
- You have until Monday to declare a partner on Canvas

Announcements

- We're not going to create pairs for A3 because there's a long list of properties which go into creating a good match, most of which we don't know
  - Programming background/interest (frontend, backend, user experience, etc.)
  - Schedule and other commitments

Announcements

- Good questions to ask a potential assignment partner:
  - When do you normally work (morning, afternoon, evening, late evening)?
  - Do you want to meet up in-person or work on the assignment remotely?
  - Are you interested in frontend development, backend development, user experience, etc.?
  - What's the minimum grade you would be comfortable receiving on this assignment?

Today's goals

By the end of today, you should be able to…

- Explain a Model-View-Controller Architecture
  and how Angular implements the architecture
- Describe the role of an Angular component
- Implement an Angular component which follows the MVC architecture

Client-side

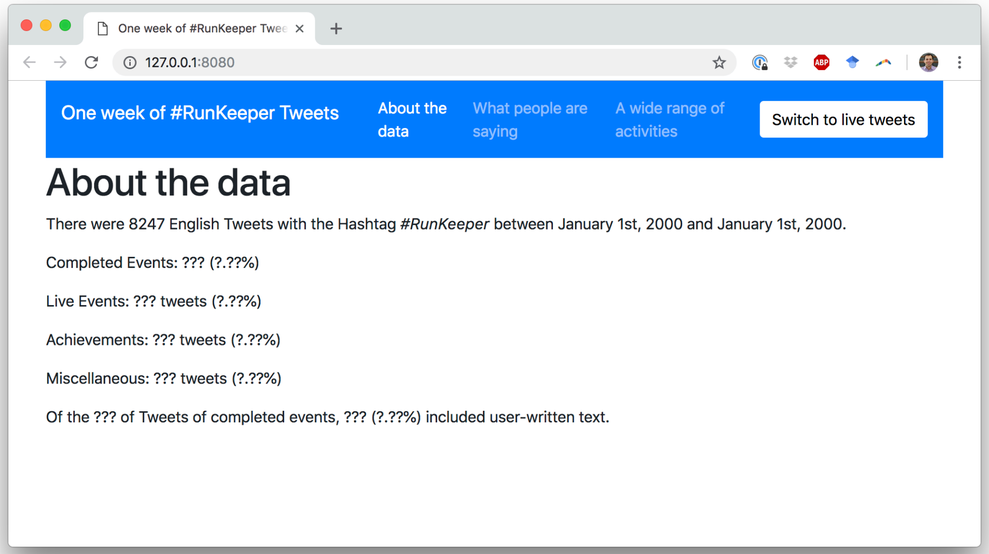
- Runs in the browser
- Changes happen in real-time in the browser
- Cannot make HTTP requests to many APIs
- Examples: AJAX, Angular, React, Vue.js

Server-side

- Runs in the command line, etc. (but maybe can still be accessed from the browser)
- Changes happen in response to HTTP requests
- Can make HTTP requests to most APIs
- Examples: Node, ASP.NET

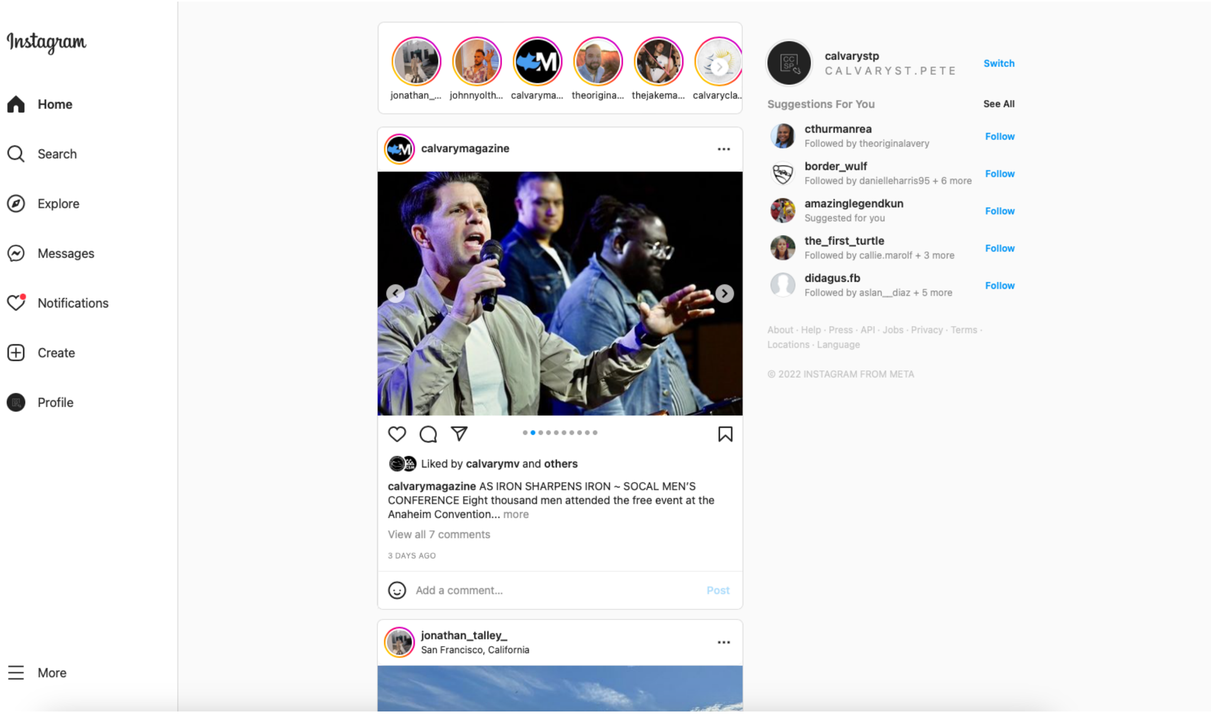
A “small” client interface

- 3 pages
- No interactivity between pages
- Data is dynamic,
  but UI is fairly static
- A lot of computation behind each page, but each page was very self-contained



A “large” client interface

- Hundreds of pages and ways to navigate between pages
- Repeated UI components (status updates)
- Different content, links, etc. displayed for each person



How do we develop large client applications?

Frameworks for large clients

- Add structure and organization
- Make UI components reusable
- Support modularity
  - Import packages, UIs, etc. when needed

Frameworks for large clients

- Angular
- React
- Vue.js
- (Insert your favorite other framework here)
- All support the same overall goal
- All have commonalities in how they function

Angular

- First released in 2009
- Uses TypeScript, HTML, and CSS
- Does not dictate what framework is used server-side
- Major releases yearly

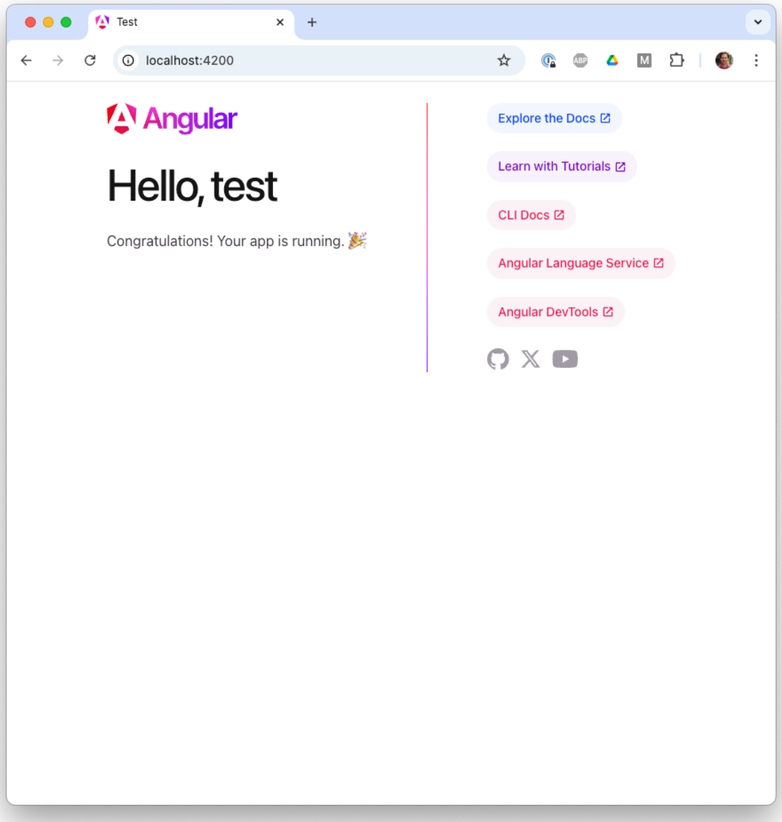
Angular != AngularJS

- Major rewrite in 2016
  - Move to TypeScript from JavaScript
  - Not backwards-compatible
- Recent syntax changes in 2024
- Makes searching StackOverflow a bit of a pain
  - But the syntax maps over reasonably okay

ANGULARJS

Angular installation

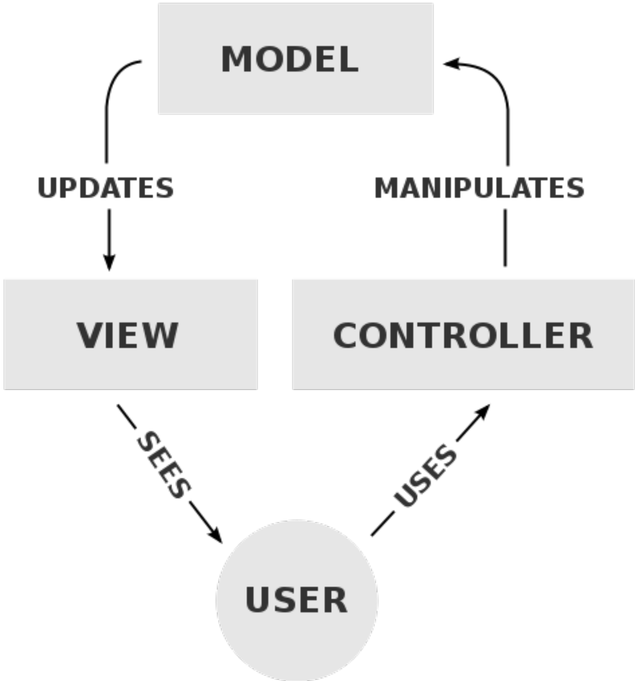
- `npm install -g @angular/cli`
- Create a new app with name `example`
  - `ng new example`
- Run app
  - `cd example`
  - `ng serve (--open)`
  - (Runs on localhost:4200 by default)



Angular architecture

Model-View-Controller

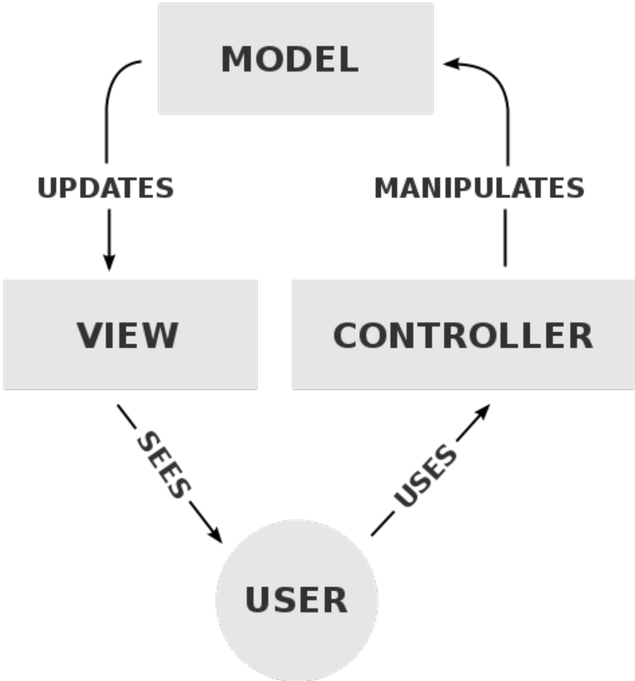
- Approach for structuring the code behind interfaces
- Model: the data behind an app
- View: the visual interface of an app
- Controller: the interaction with an app



https://en.wikipedia.org/wiki/Model%E2%80%93view%E2%80%93controller

Model-View-Controller

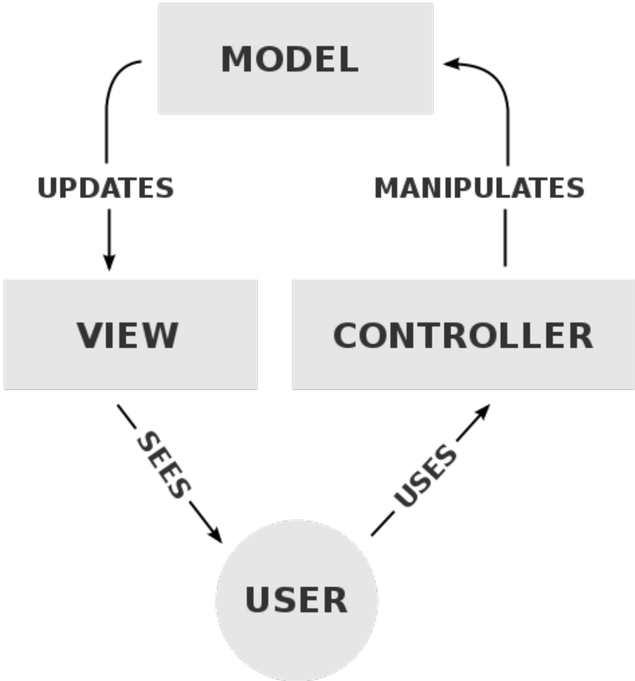
- Model: the data behind an app
  - Notifies views when it changes
  - Enables views to query the model for data
  - Allows the controller to manipulate data in the model



https://en.wikipedia.org/wiki/Model%E2%80%93view%E2%80%93controller

Model-View-Controller

- View: the visual interface of an app
  - Renders the contents of the model
  - Specifies how the model data should be presented
  - When the model changes, the view must update it's presentation
  - "Push" approach: the view waits for change notifications (live updating feed)
  - "Pull" approach: the view must ask when it wants new data (pull to refresh)
  - Forwards input to the controller



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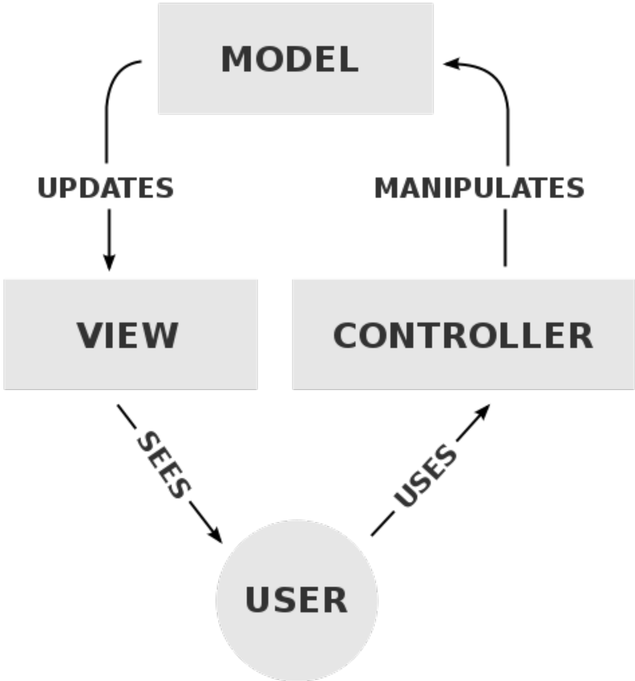
Model-View-Controller

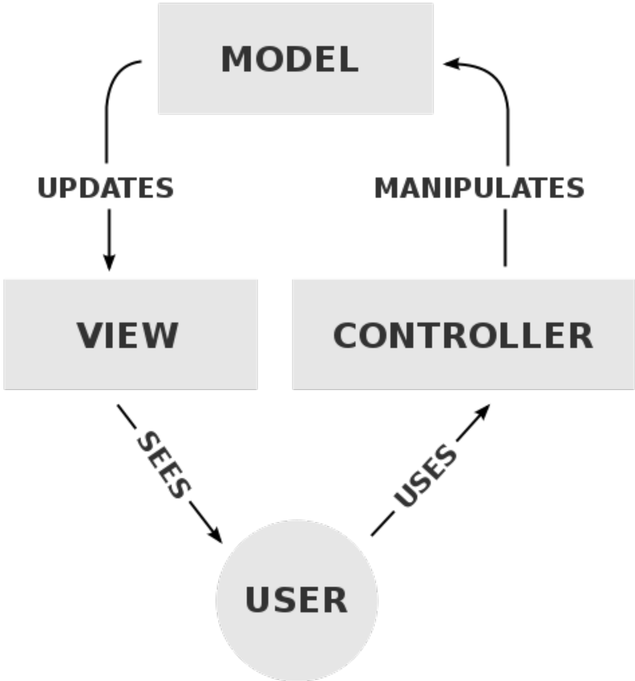
- Controller: the interaction with an app
  - Interprets user input and maps them to actions
  - Tells the model what actions to perform
  - Indirectly tells the view (through the model) if page should be rendered differently



https://en.wikipedia.org/wiki/Model%E2%80%93view%E2%80%93controller

Model-View-Controller

- Model: JavaScript for loading, parsing, and manipulating data
- View: HTML and CSS to specify layout
- Controller: event handlers for buttons and inputs in JavaScript



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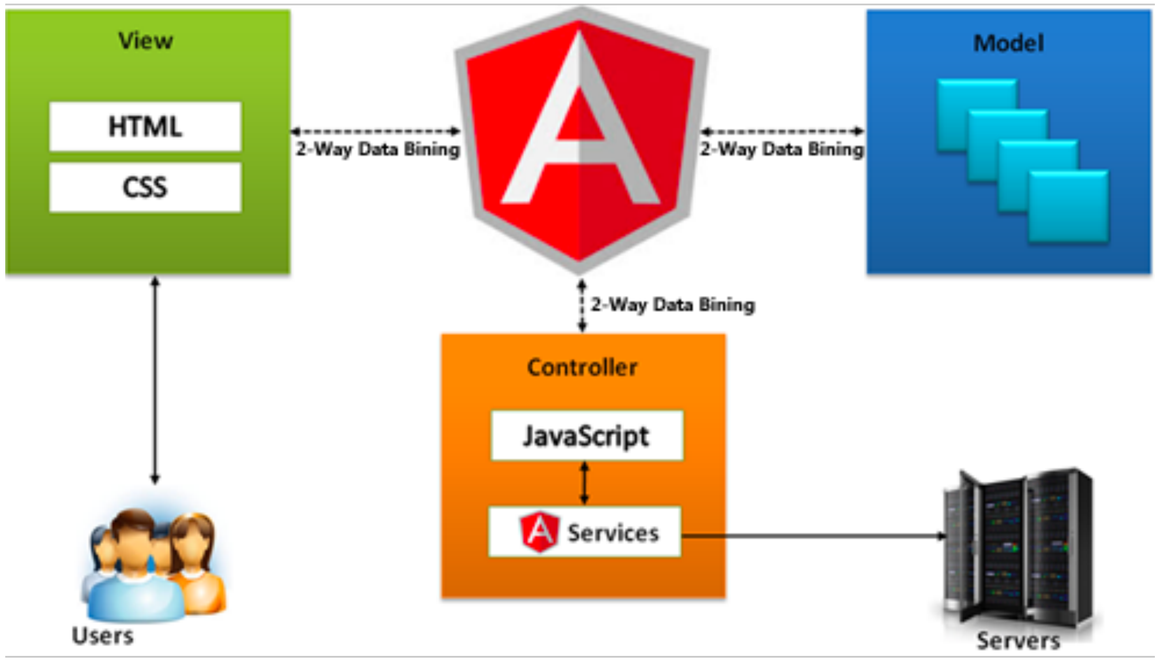
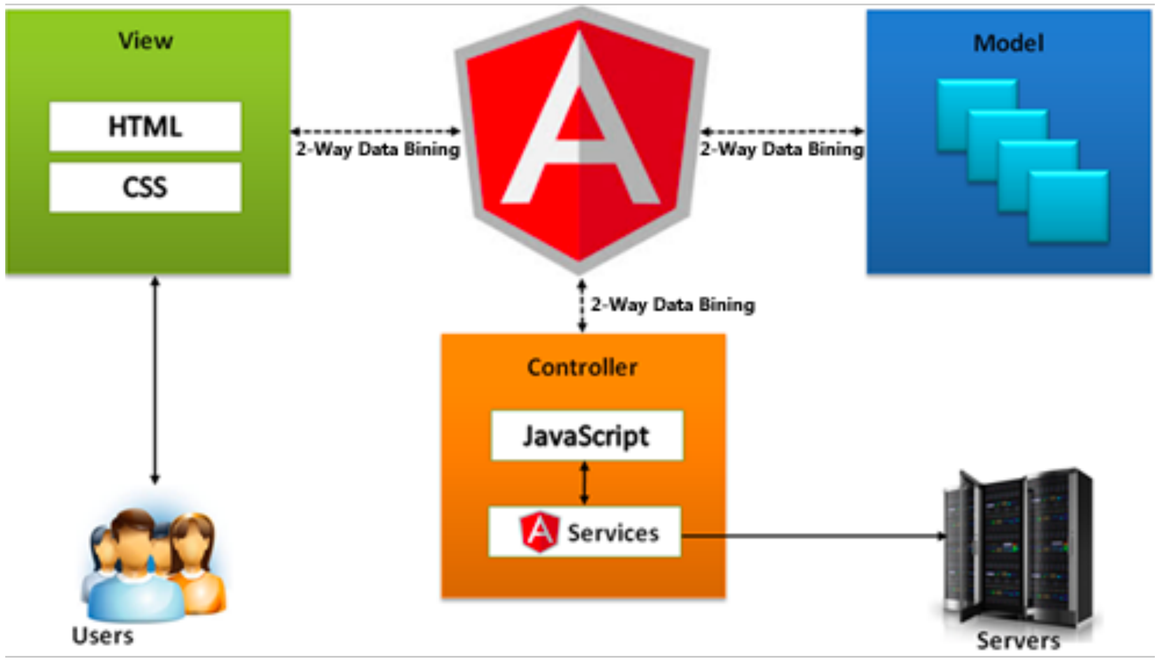
MVC in Angular

- View: HTML and CSS
- Model & Controller: TypeScript
- Angular functionality serves as the glue between the three



MVC in Angular

- **Binding**: key term
  - Variables in a view can be *bound* to variables and functions in a model or controller
  - When a variable in the *model* changes, any references to it in the *view* will also change ("push" model)
  - When a view receives input from a user, it passes it to the controller bound for that input



Following MVC in Angular

Angular components

- A component is an interface element
  - Usually larger than “a button”, but smaller than “a page”
  - Usually one which repeats across the interface

Angular components

- Component terms
  - template: the HTML file representing the view
  - style: the CSS file(s) which indicate how the component should be styled
  - selector: a CSS selector that Angular will use to all instances with this component

Angular components

- Defines the model, view, and controller for any interface element
- Make a new component: `ng generate component hello`
- Each component makes a folder consisting of four files:
  - `hello.component.css` (view)
  - `hello.component.html` (view)
  - `hello.component.spec.ts` (for automated testing; we'll mostly ignore)
  - `hello.component.ts` (model and controller)

Angular components

- Every app has at least one component
- “Root” component
  - By default, in `app.component.(html/css/ts/spec.ts)`
  - The “main” page in your app, essentially

Root template

app.component.html

```
<div>
  <h1>
    Welcome to {{ title }}!
  </h1>
  <img width="300" alt="Angular Logo" src="data:image/...">
</div>
<h2>Here are some links to help you start: </h2>
<ul>
  <li>
    <h2><a target="_blank" rel="noopener" href="https://angular.io/tutorial">Tour of Heroes</a></h2>
  </li>
  <li>
    <h2><a target="_blank" rel="noopener" href="https://github.com/angular/angular-cli/wiki">CLI
Documentation</a></h2>
  </li>
  <li>
    <h2><a target="_blank" rel="noopener" href="https://blog.angular.io/">Angular blog</a></h2>
  </li>
</ul>

<router-outlet></router-outlet>
```

Root template

app.component.html

- Looks like any other HTML page
- Only difference: page can support data binding

Root template

app.component.html

```
<div>
  <h1>
    Welcome to {{ title }}!
  </h1>
  <img width="300" alt="Angular Logo" src="data:image/...">
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<h2>Here are some links to help you start: </h2>
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  </li>
</ul>

<router-outlet></router-outlet>
```

Binds text to the variable `title`

Root component

app.component.ts

```
import { Component } from '@angular/core';
```
← Import Angular component library

```
@Component({
  selector: 'app-root',
  templateUrl: './app.component.html',
  styleUrls: ['./app.component.css']
})
export class AppComponent {
  title = 'example';
}
```

← Designates class as a component

← Replace all occurrences of `app-root` with this

← Designate template

← Designate style(s)

← All code for the component

Root component and template

app.component.html

```
<div>
  <h1>
    Welcome to {{ title }}!
  </h1>
  <img width="300" alt="Angular Logo" src="data:image/...">
</div>
<h2>Here are some links to help you start: </h2>
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  <li>
    <h2><a target="_blank" rel="noopener" href="https://
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  </li>
</ul>

<router-outlet></router-outlet>
```

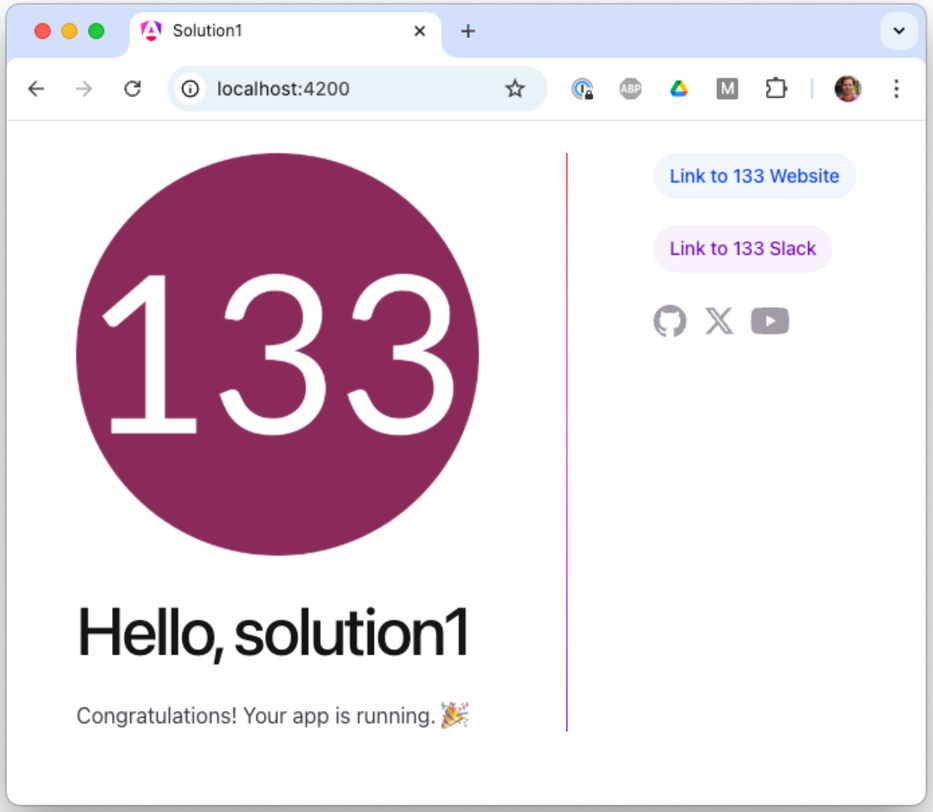
Bound to

app.component.ts

```
import { Component } from
'@angular/core';

@Component({
  selector: 'app-root',
  templateUrl: './
app.component.html',
  styleUrls: ['./
app.component.css']
})
export class AppComponent {
  title = 'example';
}
```

Hello Angular



Four types of binding

- Interpolation: {{ }}
- Property: []
- Event: ( )
- Two-way: [( )]

https://angular.io/guide/template-syntax

Interpolation binding {{ }}

- "Weave calculated strings into the text between HTML element tags and within attribute assignments"

```
<h3>
  {{title}}
  <img src="{{heroImageUrl}}" style="height:30px">
</h3>
```

- Can also be used to calculate values

```
<!-- "The sum of 1 + 1 is 2" -->
<p>The sum of 1 + 1 is {{1 + 1}}</p>
```

https://angular.io/guide/template-syntax

Property binding [ ]

- “Set an element property to a component property value”

```
<img [src]="heroImageUrl">
```

https://angular.io/guide/template-syntax

Event binding ( )

- “Listen for certain events such as keystrokes, mouse movements, clicks, and touches”

```
<!--When clicked, will run the onSave() function in
component.ts file-->
<button (click)=“onSave()">Save</button>
```

https://angular.io/guide/template-syntax

One-way binding

- Interpolation, property, and event are all one-way, or *read-only* binding
- For interpolation {{ }} and property [ ], binding goes from data source (.ts) to view target (.html)
- For event ( ), binding goes from view target (.html) to data source (.ts)

Data direction	Syntax

One-way
from data source
to view target
	`{{expression}}`
	`[target]="expression"`
	`bind-target="expression"`

One-way from view target to data source	`(target)="statement"` `on-target="statement"`
Two-way	`[(target)]="expression"` `bindon-target="expression"`

https://angular.io/guide/template-syntax

One-way binding

```
{{title}}
```

Bound to

```
<img [src]="heroImageUrl">
```

Bound to

```
<button (click)="onSave()">Save</button>
```

Bound to

```
import { Component, OnInit } from
'@angular/core';

@Component({
  selector: 'app-hello',
  templateUrl: './hello.component.html',
  styleUrls: ['./hello.component.css']
})
export class HelloComponent implements
OnInit {
  title = 'example';
  heroImageUrl = 'hero.jpg';

  onSave() {
    console.log('File saved!');
  }

}
```

https://angular.io/guide/template-syntax

Question

Which will bind the link's `href` to the link is the variable `url`?

A <a href="{{url}}">link</a>

B <a (href)="url">link</a>

C <a [href]="url">link</a>

D A and B

E A and C

Which will bind the link's `href` to the link is the variable `url`?

A <a href="{{url}}">link</a>

B <a (href)="url">link</a>

C <a [href]="url">link</a>

D A and B

E A and C

A	0%
B	0%
C	0%
D	0%
E	0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Question

Which will bind the link's `href` to the link is the variable `url`?

A <a href="{{url}}">link</a>

B <a (href)="url">link</a>

C <a [href]="url">link</a>

D A and B

E A and C

Two-way binding [( )]

- "You often want to both display a data property and update that property when the user makes changes"
- Most common use: binding to user-generated input
- ngModel directive enables two-way binding to input fields

```
<!--enteredText variable contains inputted text-->
<!--textChanged() is called after every keystroke-->
<input [(ngModel)]="enteredText" (change)="textChanged()">
```

https://angular.io/guide/template-syntax

Binding

```
<!--enteredText variable contains inputted text-->
<!--textChanged() is called after every keystroke-->
<input [(ngModel)]="enteredText" (change)="textChanged()">

<!--When clicked, will run the onSave() function in component.ts
file-->
<button (click)="onSave()">Save</button>

<h3>
  <!--will display the title-->
  {{title}}
  <!--will display the image at heroImageUrl-->
  <img [src]="heroImageUrl">
</h3>
```

Control flows

- @if and @for, with parentheses to identify what should be covered
- Switched from *ngIf and *ngFor in 2024, many online examples will use the old syntax

https://angular.love/diving-into-the-new-angular-control-flow-internals

@if

- Render a tag if condition is true

```
@if(isHalloween) {
  <p>
    Spooky!
  </p>
} @else {
  <p>
    Just another boring day in the neighborhood.
  </p>
}
```

https://angular.dev/api/core/@if

@for

- Repeat an item multiple times

```
<ul>
  @for(day of days; track day) {
    <li>
      {{day}}
    </li>
  }
</ul>
```

- Sunday
- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday

```
//In component.ts
  days =
["Sunday",
"Monday",
"Tuesday",
"Wednesday",
"Thursday",
"Friday",
"Saturday"];
```

- "Track" is for performance, should be a unique property

- Can optionally specify index

```
<ul>
  @for(day of days; track day; let i = $index) {
    <li>
      {{i+1}}: {{day}}
    </li>
  }
</ul>
```

- 1: Sunday
- 2: Monday
- 3: Tuesday
- 4: Wednesday
- 5: Thursday
- 6: Friday
- 7: Saturday

https://angular.dev/api/core/@for

Using components

- Components can import other components
  - Follow the selector defined in the component's `.ts` file
- In app.component.html:

```
<div>
  <h1>
    Welcome to {{ title }}!
  </h1>
  <app-day></app-day>
</div>

<router-outlet></router-outlet>
```

Welcome to example!

- Sunday
- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday

Using components

- Components can specify inputs

```
import { Component, OnInit, Input }
from '@angular/core';

@Component({
  selector: 'app-day',
  templateUrl: './day.component.html',
  styleUrls: ['./day.component.css']
})
export class DayComponent {
  @Input() today:string;

  days = ["Sunday", "Monday",
"Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday"];
}
```

Input

```
<ul>
@for(let day of days){
  <li>
    {{day}}
      @if(day == today) {
        <strong>Today!</strong>
      }
  </li>
}
</ul>
```

Input referenced in .html

Using components

- Inputs are then passed:
  - As properties if they're dynamic
  - Like any other attribute if they're static

```
<div>
  <h1>
    Welcome to {{ title }}!
  </h1>
  <app-day [today]="dayOfWeek"></app-day>
</div>
```

Sets `day` property to `dayOfWeek` variable

```
<div>
  <h1>
    Welcome to {{ title }}!
  </h1>
  <app-day today="Friday"></app-day>
</div>
```

Sets `day` property static value `Friday`

Using components

- Can also specify output properties

```
@Output('myClick') clicks = new EventEmitter<string>();
```

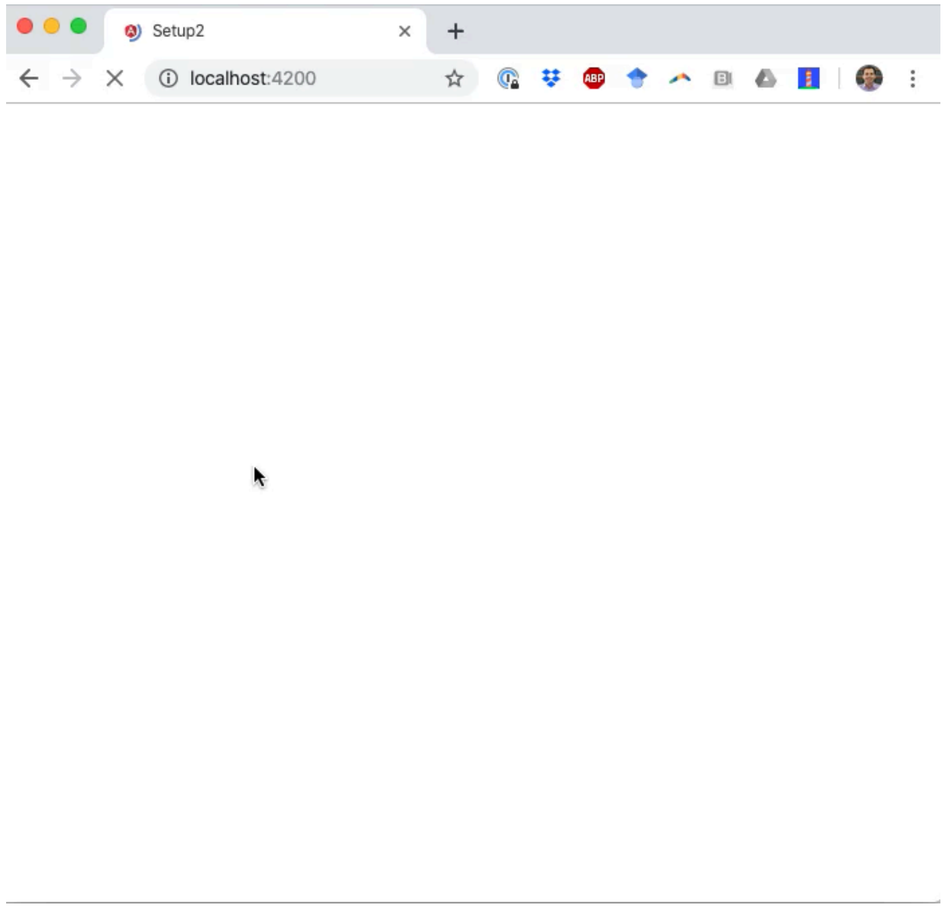
- When adding component, can specify an event to trigger when `clicks()` is called

```
<app-button (myClick)="clickMessage">click with myClick</
app-button>
```

- The event will be triggered in the parent component

```
clickMessage() {
 console.log("clicked!");
}
```

Rolling Dice



Summary

- Angular is made up of components, which are UI elements which each follow a Model-View-Controller framework
  - `.html` and `.css` file define the view
  - `.ts` defines the model and controller
- Data moves between model and view through binding
  - Components can pass data by binding `Input`s and `Output`s
- Components can include other components
  - `@if` and `@for` help determine whether and how many components to create

Today's goals

By the end of today, you should be able to…

- Explain a Model-View-Controller Architecture
  and how Angular implements the architecture
- Describe the role of an Angular component
- Implement an Angular component which follows the MVC architecture

IN4MATX 133: User Interface Software

Lecture 10:
Components in Angular